

DOCUMENT 04

Carrier Operations & Logistics Model

Launch Sequencing · Battery Swap Protocol · Multi-Carrier · Energy Budgets

Complete engineering specification for K1 ground and K2 airborne carrier operations. Covers hardware design, battery swap step-by-step procedure, worker-carrier rendezvous algorithms, failure handling, energy budget analysis, and multi-carrier coordination for missions beyond single-carrier endurance.

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1. WHY A CARRIER IS THE CRITICAL PATH

The 7+ hour continuous fleet coverage claim in the core paper is conditional on the existence of a functioning carrier platform. The fleet endurance scheduling model ($N_w \times t_f / t_r$ workers continuously airborne) is mathematically sound, but its physical realisation requires a carrier that can: (a) hold at least two docking bays simultaneously, (b) recharge worker batteries at $\geq 2C$ rate within 30 minutes, and (c) carry sufficient reserve battery capacity for the full mission duration. This document provides the full engineering specification for both carrier variants and the operational protocols that turn the scheduling model into demonstrated capability.

2. K1 GROUND CARRIER — ENGINEERING SPECIFICATION

2.1 System Architecture

K1 is a ruggedised IP65-rated ground vehicle with two integrated D2 docking bays, a six-pack battery management bank, communication relay electronics, and differential drive motors for autonomous repositioning within a 500 m deployment radius. It operates on mains power via a 50 m weatherproof cable for unlimited duration, or on 2× 24V 20 Ah LiFePO4 internal packs (960 Wh total) for up to 4 hours cable-free. The electronics bay is climate-controlled (heater for <5°C ambient) to maintain battery performance and prevent condensation on charging electronics.

Subsystem	Specification	Key Components	Notes
Chassis	800×600×400 mm Al extrusion, IP65	Aluminium extrusion, EPDM gaskets	Hinged lid; cable glands for all penetrations
Drive	4× DC brushed 12V 80W, differential steer	Pololu 37D gearmotors, 1:20 ratio, encoders	Max speed 0.5 m/s; ≥30 kg payload capacity
Drive control	Dual-channel brushed controller	Roboteq MDC2230 60A CAN	Closed-loop velocity control via encoders
Self-navigation	GPS waypoint + compass + odometry	u-blox M9N, HMC5983, encoder fusion	3 m absolute position accuracy
Dock bay ×2	D2 module, integrated IR+pogo+mag	See D2 specification	Side-by-side; 200 mm separation centre-centre
Charging ×2	CC-CV Li-ion charger per bay	Meanwell RSP-320-29.2 (29.2V/11A)	321 W per bay = 2.15C for 150 Wh battery
Battery bank	6× worker battery slots, active BMS	6× Daly 4S 100A BMS modules	Hot-swap storage; charged ready-to-go
Onboard power	2× LiFePO4 24V 20Ah	CATL prismatic cells	960 Wh; supports ~4 complete worker recharge cycles
Communication relay	900 MHz LoRa 500 mW + ESP32 mesh hub	Holybro SiK 500mW + ESP32-S3	Extends fleet comms range to 8–10 km from GCS
Control MCU	STM32H743 on custom carrier board	Custom PCB: drive + dock + BMS + comms	Manages all K1 subsystems via CAN + UART
Climate control	Heater 100W + thermostat; fan 20W	PTC heater; temp sensor NTC10K	Maintain bay temp 15–45°C for optimal recharge
Weatherproofing	IP65 sealed enclosure; IP54 connectors	Amphenol industrial	Operational in rain, dust, 0–50°C ambient

Table 1. K1 ground carrier subsystem specifications.

2.2 K1 Energy Budget — Detailed Analysis

For a 7-hour mission with 5 workers and 30-minute rotation cycles, K1 must supply:

Number of recharge cycles: 7 hours × (60 min/hr) / 30 min cycle = 14 recharge events

Energy per recharge (150 Wh at 90% charger efficiency): 167 Wh/event

Total recharge energy: 14 × 167 = 2,338 Wh

K1 self-consumption (drive 40W + electronics 30W + climate 20W): 90W × 7h = 630 Wh

Total K1 energy required: 2,338 + 630 = 2,968 Wh

K1 onboard capacity: 960 Wh — covers only 32% of 7-hour mission

Conclusion: K1 on-board batteries alone support approximately 2.3 hours of sustained operations. For full 7-hour missions, K1 must be connected to mains power. For cable-free deployments beyond 4 hours, use K2 airborne carrier or deploy 2× K1 units alternating between mains-connected and field-deployed positions.

2.3 K1 BOM and Cost

Category	Items	Subtotal Cost
Chassis + mechanical	Al extrusion, fasteners, wheels, hinges	\$320
Drive system	4× Pololu gearmotors + Roboteq MDC2230	\$460
Navigation	M9N GPS + IMU + compass + wiring	\$120
Dock bays ×2	2× D2 modules (see D2 spec)	\$440
Charging electronics	2× Meanwell RSP-320-29.2 chargers	\$170
BMS modules ×6	6× Daly 100A 4S BMS	\$150
Onboard batteries	2× 24V 20Ah LiFePO4	\$400
Communication relay	LoRa 500mW + ESP32-S3 + antennas	\$95
Control MCU + PCB	Custom STM32H7 carrier controller board	\$110
Climate control	PTC heater + fan + thermostat	\$65
Wiring + connectors	Marine IP65 connectors, cable glands, harness	\$90
Mains PSU (external)	Meanwell SDR-960-24 (960W 24V)	\$180
	TOTAL K1	~\$2,600

Table 2. K1 ground carrier BOM and cost.

3. K2 AIRBORNE VTOL CARRIER — SPECIFICATION

3.1 Design Philosophy

K2 extends the operational range envelope to 5 km from the ground station by flying to a position within the mission area and hovering as a mobile charging and coordination platform. Unlike K1 which has unlimited energy via mains connection, K2 is energy-constrained by its own flight battery. The fundamental K2 design tension: carrying more worker batteries increases K2 mass, increases K2 hover power, and reduces K2 endurance. The optimal worker battery payload is derived from the same $W^{1.5}$ framework:

$P_{K2} = k \times (M_{K2} \times g)^{1.5}$ where $k = 0.639 \text{ W/N}^{\{3/2\}}$

At $M_{K2} = 5 \text{ kg}$ (MTOW): $P_{\text{hover}} \approx 0.639 \times (5 \times 9.81)^{1.5} = 0.639 \times 344 = 220 \text{ W}$

K2 flight battery: 450 Wh (6S 10,000 mAh Li-ion). Usable at 80% DoD: 360 Wh.

K2 hover endurance (no recharge load): $360/220 \times 60 = 98 \text{ min}$

K2 endurance under 2× 100W recharge load: $360/(220+200) \times 60 = 51 \text{ min}$

Practical K2 sortie: 40 min in field (with safety margin); return before 100% discharge

3.2 K2 Worker Rendezvous Protocol

K2 airborne docking introduces additional challenges vs K1 ground docking: K2 itself has position uncertainty of $\pm 0.5\text{--}1.0\text{ m}$ in 8 m/s wind (5 kg VTOL RTK GPS hold). The worker must dock to a moving target within this uncertainty. The two-phase protocol:

- **Phase A — GPS Rendezvous:** Worker navigates to virtual waypoint 30 m directly above K2 GPS position. K2 broadcasts its current GPS position at 5 Hz on ESP-NOW mesh.
- **Phase B — IR Beacon Guided:** Within 30 m, worker switches from GPS position control to IR beacon error signal. K2 IR beacons transmit continuously; worker locks on and descends along ϵ error signal.
- **Timing constraint:** K2 must hold near-stationary hover during final approach ($\pm 0.5\text{ m}$). K2 flight controller enters precision hover mode (reduced gain, high-frequency position hold) triggered by DOCK_APPROACH message from worker.
- **Weather limit:** K2 airborne docking is suspended at wind speeds $>8\text{ m/s}$ at the K2 altitude. K2 descends to ground and K1 protocol resumes.
- **Latency:** Total time from CBBA-SOC dock assignment to worker docked and charging: $\sim 8\text{ min}$ (3 min transit at 5 m/s + 2 min IR approach + 3 min alignment). K2 holds hover for this 8-min window.

Phase	Worker Navigation	K2 State	Guidance Signal	Altitude
1: Transit (0–3 min)	GPS waypoint to 50 m above K2	Normal mission or loiter	GPS + ESP-NOW K2 position	$>50\text{ m AGL}$
2: Approach (3–5 min)	GPS position hold at 30 m above K2	Precision hover mode	K2 GPS + worker relative error	$\sim 30\text{ m above K2}$
3: IR lock (5–7 min)	IR beacon error signal primary	Stationary precision hold	IR beacon differential	2 m above dock
4: Contact (7–8 min)	Slow descent 0.3 m/s on IR error	Hold position + open dock	IR error $\rightarrow$ pogo contact	0 (docked)
5: Charging (8–38 min)	Docked — motors off	Resume mission or loiter	None (docked)	0 (on K2)

Table 3. K2 airborne docking protocol phases.

4. BATTERY SWAP — DETAILED STEP-BY-STEP PROCEDURE

The following 8-step procedure governs every battery rotation event on both K1 and K2:

Step 1 — SOC Monitoring	CBBA-SOC continuously monitors all worker SOC values at 1 Hz. When any worker falls below SOC_dock threshold (35% default, configurable), a DOCK_RETURN task is created in the scheduler with bid priority inversely proportional to remaining SOC.
Step 2 — Assignment	The worker accepts the DOCK_RETURN task if its CBBA-SOC bid wins against all other available tasks. The scheduler simultaneously reassigns the departing worker coverage area to remaining active workers.
Step 3 — Departure Announcement	Worker broadcasts WORKER_DEPART {worker_id, dock_id, eta_seconds} on ESP-NOW mesh at 1 Hz. Command node updates fleet state and GCS map.
Step 4 — Bay Reservation	Carrier controller assigns an available dock bay to the inbound worker. If both bays occupied, assigns a time slot (worker orbits at safe altitude). Bay status LED (visible to operator) changes from GREEN to AMBER (reserved).
Step 5 — Approach	Worker navigates to carrier via GPS (phase A) then IR beacon (phase B). Final approach speed: $<0.5\text{ m/s}$ horizontal, $<0.3\text{ m/s}$ vertical within 2 m of dock.

Step 6 — Magnetic Capture	Worker belly keel enters funnel guide. N52 magnets (64 N total holding force) capture worker within ±50 mm lateral, ±20 mm vertical. Carrier SSR detects pogo-pin resistance signature (unique per worker; prevents accidental activation on non-drone contacts). SSR closes; charging begins at 4A CC.
Step 7 — Recharge	Charger executes CC-CV profile: 4A constant current until cell voltage reaches 16.4V (4.1V/cell × 4), then CV hold until current drops to 0.2A. Total time 0–80% SOC: ≤25 min. BMS monitors: cell temperatures, cell balance (±20 mV target), total pack voltage. If any cell exceeds 60°C, charging rate halved automatically.
Step 8 — Launch Clearance	When SOC ≥ 80% AND CBBA-SOC issues LAUNCH command AND operator has not issued HOLD: (a) SSR opens (power disconnected), (b) magnetic latch releases via vibration motor pulse on dock, (c) worker arms motors and ascends to 5 m AGL, (d) worker rejoins ESP-NOW mesh and CBBA-SOC assigns next task, (e) bay status LED returns GREEN.

5. MULTI-CARRIER COORDINATION

Scenario	Configuration	Continuous Coverage	Limiting Factor	Phase
Ground single carrier	1× K1 on mains + 3 workers	7+ hours	Coverage radius 500 m	1
Airborne single carrier	1× K2 + 5 workers	~4 hours (K2 self-limited)	K2 must land and recharge	2
Dual airborne relay	2× K2 alternating + 5 workers	12+ hours	K2 ground recharge time ~50 min	2
Ground+Air hybrid	1× K1 (near base) + 1× K2 (forward) + 5 workers	24+ hours	K2 recharges on K1; indefinite	3
Persistent (triple carrier)	1× K1 base + 2× K2 relay + 8 workers	Indefinite	Ground power supply only	3

Table 4. Multi-carrier fleet configurations and endurance.

5.1 K2 Recharging on K1

K2 can recharge its own flight battery on K1 by landing on a dedicated K2 dock pad mounted on K1's upper surface. K2 dock pad specification: 300×300 mm aluminium plate; 6× XT90 contacts for 48V/20A = 960W charge rate. K2 flight battery (450 Wh) charges from 20% to 80% in: $(450 \times 0.60 / 960 \times 60) = 17$ minutes at 960W.

This enables a K1+K2 relay cycle: K2 operates in field for 40 minutes, lands on K1 for 20-minute fast recharge, returns to field. Net K2 availability: $40/(40+20) = 67\%$ of mission time, sufficient for continuous fleet coverage with 5 workers.

6. CARRIER VALIDATION TEST PLAN

Test	Setup	Metric	n	Pass Criterion	Timeline
K1 static dock bench	Worker (motors off) lands on K1; measure alignment	Alignment error, contact resistance	50	≥95% success; error ≤20 mm	Month 4–5
K1 hover-to-dock	Worker hovers → autonomous land on K1 in 0 m/s wind	Success rate, time	20	≥90% success; ≤8 min cycle	Month 5–6

Test	Setup	Metric	n	Pass Criterion	Timeline
K1 wind-stress dock	Same as above in 4 m/s and 8 m/s wind	Success rate vs wind	20 per condition	≥85% at 8 m/s	Month 6–7
K1 rotation cycle	2 workers; full 2-hour rotation on K1+mains	Battery failure events, coverage continuity	5 runs	Zero battery failures; cycle ≤35 min	Month 7–8
K1 4-hour endurance	K1 on internal battery (no mains); 2 workers	K1 SOC over time; recharge cycles completed	2 runs	≥3 complete rotation cycles; K1 SOC ≥20% at end	Month 8–9
K2 airborne dock (tethered)	K2 hovers 5 m AGL on tether; worker attempts dock	Success rate	20	≥70% success (Phase 2 stretch)	Month 12–15

Table 5. Carrier validation test plan.